Northwind Catalog – Week 15 Assessment
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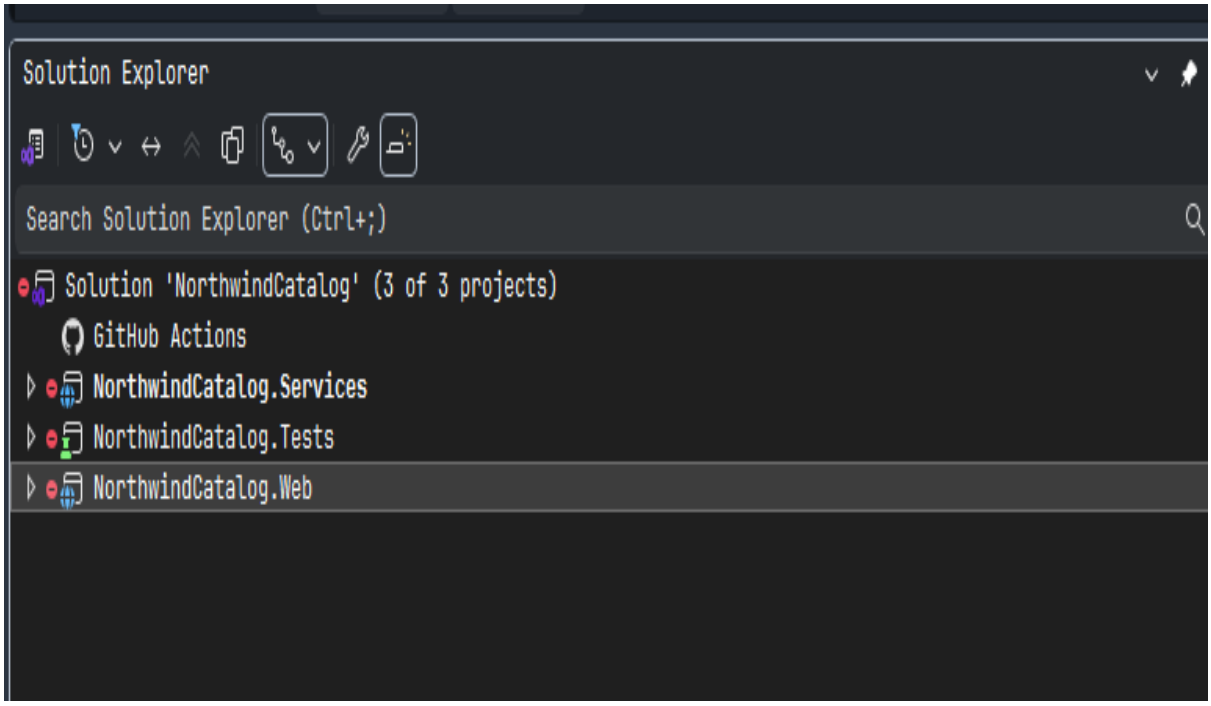
The solution consists of three projects:

- NorthwindCatalog.Services → Web API + Business Logic + EF Core
- NorthwindCatalog.Web → MVC UI (API Consumption)
- NorthwindCatalog.Tests → Unit Testing using xUnit



Fluent Api for scaffolding:

Use following commands in Package Manager:

Scaffold-DbContext "Connection_String" Microsoft.EntityFrameworkCore.SqlServer -OutputDir Models -Tables Categories,Products

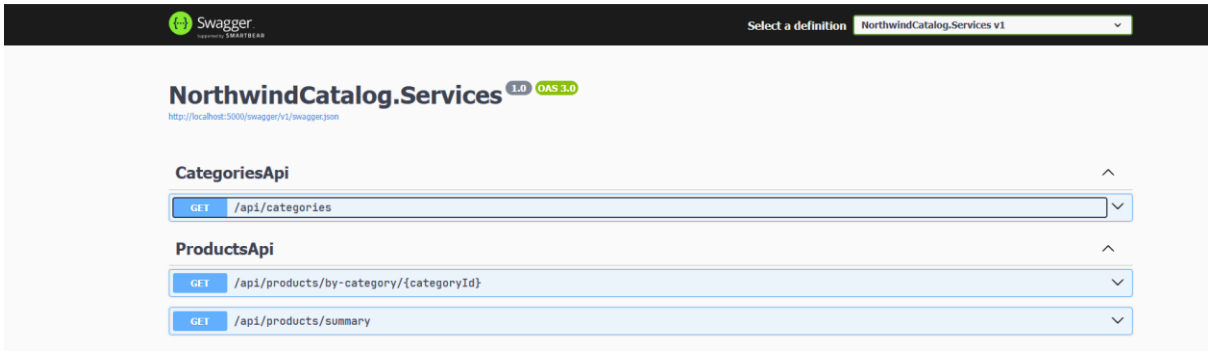
The connection string will be present in the appsettings.json file.

```
    protected override void OnConfiguring(DbContextOptionsBuilder optionsBuilder)
    => optionsBuilder.UseSqlServer("Server=localhost\\SQLEXPRESS02;Database=Northwind;Trusted_Connection=True;TrustServerCertific

    0 references
    protected override void OnModelCreating(ModelBuilder modelBuilder)
    {
        modelBuilder.Entity<Category>(entity =>
        {
            entity.HasIndex(e => e.CategoryName, "CategoryName");

            entity.Property(e => e.CategoryId).HasColumnName("CategoryID");
            entity.Property(e => e.CategoryName).HasMaxLength(15);
            entity.Property(e => e.Description).HasColumnType("ntext");
            entity.Property(e => e.Picture).HasColumnType("image");
        });

        modelBuilder.Entity<Product>(entity =>
        {
            entity.HasIndex(e => e.CategoryId, "CategoriesProducts");

            entity.HasIndex(e => e.CategoryId, "CategoryID");

            entity.HasIndex(e => e.ProductName, "ProductName");

            entity.HasIndex(e => e.SupplierId, "SupplierID");

            entity.HasIndex(e => e.SupplierId, "SuppliersProducts");

            entity.Property(e => e.ProductId).HasColumnName("ProductID");
            entity.Property(e => e.CategoryId).HasColumnName("CategoryID");
            entity.Property(e => e.ProductName).HasMaxLength(40);
            entity.Property(e => e.QuantityPerUnit).HasMaxLength(20);
            entity.Property(e => e.ReorderLevel).HasDefaultValue((short)0);
            entity.Property(e => e.SupplierId).HasColumnName("SupplierID");
            entity.Property(e => e.UnitPrice)
                .HasDefaultValue(0m)
                .HasColumnType("money");
            entity.Property(e => e.UnitsInStock).HasDefaultValue((short)0);
            entity.Property(e => e.UnitsOnOrder).HasDefaultValue((short)0);

            entity.HasOne(d => d.Category).WithMany(p => p.Products)
                .HasForeignKey(d => d.CategoryId)
                .HasConstraintName("FK_Products_Categories");
        });

        OnModelCreatingPartial(modelBuilder);
    }

    1 reference
    partial void OnModelCreatingPartial(ModelBuilder modelBuilder);
}
```

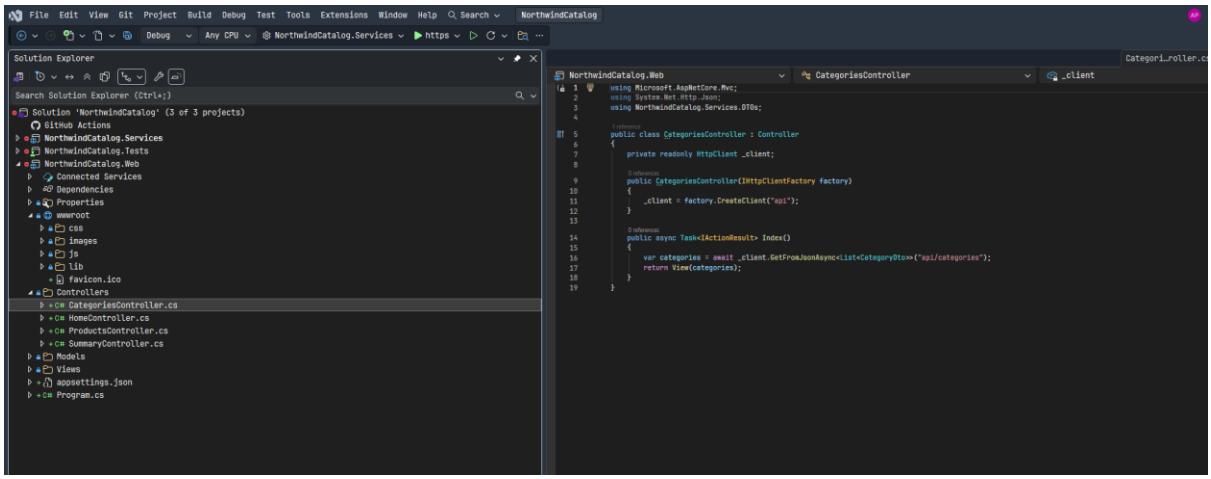
Services Project Folder structure (Repository Based):

```
Solution 'NorthwindCatalog' (3 of 3 projects)
    GitHub Actions
    NorthwindCatalog.Services
        Connected Services
        Dependencies
        Properties
            launchSettings.json
        Controllers
            CategoryController.cs
            ProductController.cs
            WeatherForecastController.cs
        Data
            NorthwindContext.cs
        DTOs
            CategoryDto.cs
            CategorySummaryDto.cs
            ProductDto.cs
        Mapping
            MappingProfile.cs
        Models
            Category.cs
            Product.cs
        Repository
            CategoryRepository.cs
            ICategoryRepository.cs
            IProductRepository.cs
            ProductRepository.cs
        appsettings.json
        NorthwindCatalog.Services.http
        Program.cs
        WeatherForecast.cs
    NorthwindCatalog.Tests
    NorthwindCatalog.Web
```

Services API Swagger:

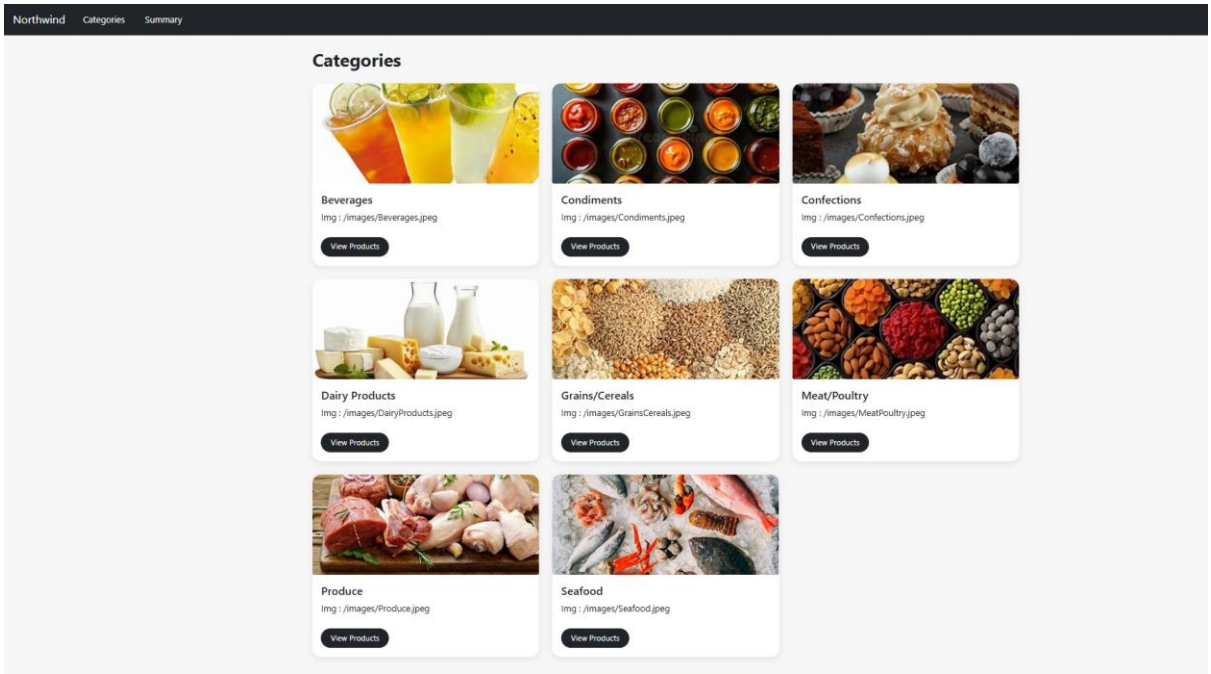


MVC Controllers:



- Save the images in the wwwroot/images folder so that it can be used by the automapper.

Categories Page:



Product Detail Page:

Products

Chai
Price: ₹18.0000
Stock: 39
Inventory: ₹702.0000

Chang
Price: ₹19.0000
Stock: 17
Inventory: ₹323.0000

Guaraná Fantástica
Price: ₹4.5000
Stock: 20
Inventory: ₹90.0000

Sasquatch Ale
Price: ₹14.0000
Stock: 111
Inventory: ₹1554.0000

Steeleye Stout
Price: ₹18.0000
Stock: 20
Inventory: ₹360.0000

Côte de Blaye
Price: ₹263.5000
Stock: 17
Inventory: ₹4479.5000

Chartreuse verte
Price: ₹18.0000
Stock: 69
Inventory: ₹1242.0000

Ipoh Coffee
Price: ₹46.0000
Stock: 17
Inventory: ₹782.0000

Laughing Lumberjack Lager
Price: ₹14.0000
Stock: 52
Inventory: ₹728.0000

Outback Lager
Price: ₹15.0000
Stock: 15
Inventory: ₹225.0000

Rhönbräu Klosterbier
Price: ₹7.7500
Stock: 125
Inventory: ₹968.7500

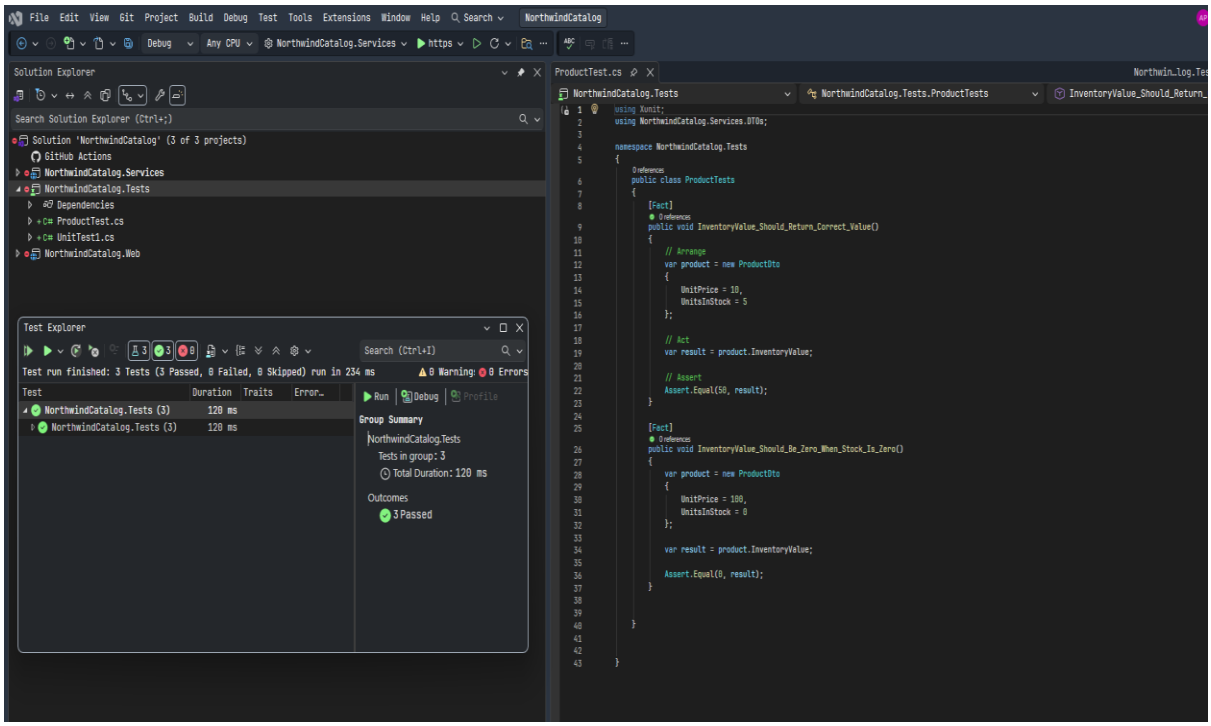
Lakkalikööri
Price: ₹18.0000
Stock: 57
Inventory: ₹1026.0000

Summary Page:

Summary

Category	Products	Avg Price	Most Expensive
Beverages	12	₹37.979166	Côte de Blaye
Condiments	12	₹23.062500	Vegie-spread
Confections	13	₹25.160000	Sir Rodney's Marmalade
Dairy Products	10	₹28.730000	Raclette Courdavault
Grains/Cereals	7	₹20.250000	Gnocchi di nonna Alice
Meat/Poultry	6	₹54.006666	Thüringer Rostbratwurst
Produce	5	₹32.370000	Manjimup Dried Apples
Seafood	12	₹20.682500	Carnarvon Tigers

Testing:


File Edit View Git Project Build Debug Test Tools Extensions Window Help Search NorthwindCatalog
Debug Any CPU NorthwindCatalog.Services https
Solution Explorer
Search Solution Explorer (Ctrl+;)
Solution 'NorthwindCatalog' (3 of 3 projects)
GitHub Actions
NorthwindCatalog.Services
NorthwindCatalog.Tests
Dependencies
ProductTest.cs
UnitTest1.cs
NorthwindCatalog.Web
Test Explorer
Search (Ctrl+I)
Test run finished: 3 Tests (3 Passed, 0 Failed, 0 Skipped) run in 234 ms
0 Warning 0 Errors
Test Duration Traits Error…
NorthwindCatalog.Tests (3) 120 ms
NorthwindCatalog.Tests (3) 120 ms
Run Debug Profile
Group Summary
NorthwindCatalog.Tests
Tests in group: 3
Total Duration: 120 ms
Outcomes
3 Passed
ProductTest.cs
NorthwindCatalog.Tests
NorthwindCatalog.Tests.ProductTests
using Xunit;
using NorthwindCatalog.Services.DTOs;
namespace NorthwindCatalog.Tests
{
0 references
public class ProductTests
{
[Fact]
0 references
public void InventoryValue_Should_Return_Correct_Value()
{
// Arrange
var product = new ProductDto
{
UnitPrice = 10,
UnitsInStock = 5
};
// Act
var result = product.InventoryValue;
// Assert
Assert.Equal(50, result);
}
[Fact]
0 references
public void InventoryValue_Should_Be_Zero_When_Stock_Is_Zero()
{
var product = new ProductDto
{
UnitPrice = 100,
UnitsInStock = 0
};
var result = product.InventoryValue;
Assert.Equal(0, result);
}
}
}